

MFNavis User Manual LCD and Web

Draft user manual · October 9, 2026

MFNavis identifies stars in camera images to show where your telescope is pointing and guide you toward an observing target. This manual explains how to prepare for observing, find objects, and change settings using the LCD, keypad, and web interface. A connected INDI mount also supports automatic and manual movement. **INDI MOUNT connection and detailed settings are available on the web server's INDI page.** See [chapter 14](#) for connecting to the web interface and using each page.

Tables, instructions, and menu paths show **official English menu names (Korean UI labels)** together, for example Start(시작), Focus(초점), and Set Filters(필터 설정). Names such as Lens, Distortion, and CALIB also appear in English in the Korean UI, so they are written once. Parentheses contain the actual Korean UI label. The menu diagrams explain menu relationships. LCD captures show the current UI at 128×128 resolution; location, time, measurements, and mount status are examples for the procedures. Actual text size and layout depend on your device. INDI menus appear **only when Mount Control(가대 제어) is enabled.**

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1 Quick start for your first observing session

Follow these steps for your first session. **Plate solving** identifies the stars in a camera image to determine the current pointing direction. **Alignment** matches the camera's reference direction to the center of your eyepiece view.

MFNavis / Observing sequence

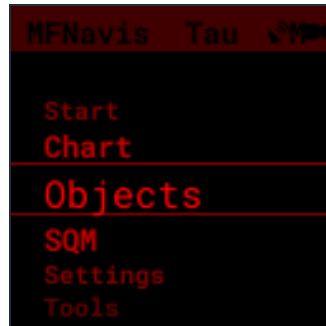


1. **Focus the camera:** Open `Start(시작) → Focus(초점)`. Adjust the MFNavis camera lens while viewing a bright star, looking for the position with a smaller HFD value. This adjustment is for the camera lens.
2. **Check location and time:** Open `Start(시작) → GPS Status(GPS 상태)` and check that a location is available. If GPS is unavailable, use `Tools(도구) → Place & Time(위치/시간)` to enter your location first, then the time and date.
3. **Align the observing center:** Center a bright star in the telescope eyepiece and open `Start(시작) → Align(정렬)`. Press `□` to start star selection, use the direction keys to select the same star, then press `□` again.
4. **Choose an object:** Open `Objects(천체) → By Catalog(카탈로그별) → Messier`. Select an object with `↑ / ↓` and press `→` to open its details.
5. **Find the object:** Move the telescope using the direction indicators and remaining angle in the details screen. For automatic movement, check the mount connection first, then press `5` in object details. After GoTo completes and the mount tracks the selected object, a rectangular border appears around the Push view.
6. **Log the observation:** From object details, press `→` to open `LOG(로그)` and enter your ratings. Select the save entry and press `→` to save.
7. **Shut down:** Select `Tools(도구) → Power(전원) → Shutdown(종료) → Confirm(확인)`. Wait for shutdown to finish before switching off power.

If you lose your place in the menus: Hold `←` to return to the top-level MFNavis menu.

2 The menu structure at a glance

There are six top-level menus. Use **Start(시작)** before observing, **Chart(성도)** and **Objects(천체)** to find objects, **Settings(설정)** for display and device settings, and **Tools(도구)** to check status and shut down.



MFNavis main menu · the selected entry is centered

MFNavis / Menu overview



* Visible only when Mount Control is On. Child lists are abbreviated.

Menu	Purpose	Typical first-use path
Start(시작)	Focus, align camera and eyepiece centers, check GPS, connect INDI	Start(시작) → Focus(초점)
Chart(성도)	Show the sky chart for the current pointing direction	Chart(성도)
Objects(천체)	Object catalogs, search, observing lists, and filters	Objects(천체) → By Catalog(카탈로그별) → Messier
SQM	Measure the brightness of the sky seen by the camera	SQM
Settings(설정)	Display, camera, communications, INDI, and hardware settings	Settings(설정) → User Pref...(사용자...)
Tools(도구)	Status, equipment, location and time, updates, and power	Tools(도구) → Status(상태)

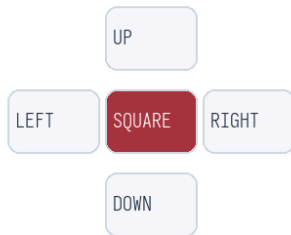
Reading a path: Objects(천체) → By Catalog(카탈로그별) → Messier means select Objects(천체) and press → , select By Catalog(카탈로그별) and press → , then select Messier and press → . Follow menu names even if their positions change.

3 Common controls

3.1 Buttons and press types

In this manual,  means the keypad's **SQUARE** button. A short press means press and release once. A long press means hold until the long-press action runs.

MFNavis / Common controls (schematic)



UP / DOWN: choose an item
RIGHT: open / select LEFT: go back
Hold LEFT: return to MFNavis menu
Hold SQUARE: Quick Menu
SQUARE + PLUS / MINUS: LCD brightness

Button	Action in list menus	Other screens
↑ / ↓	Select previous / next entry	Adjust exposure in Focus(초점); select stars or reference points during alignment
→	Open an entry, select a value, or execute a command	Advance or confirm in entry screens; open the observation log from object details
←	Go back one level	May move to a previous field or move a reference point during entry or alignment
Hold ←	Return to the top-level menu	Check the screen's cancellation procedure if a task is in progress
Hold →	Open details of the most recently viewed object	Requires a recent object; does not open a new screen while already in object details
Short press 	Depends on the active screen	Switch views or start / confirm alignment. Use → to select ordinary menu entries
Hold 	Open / close the current screen's quick menu	Available on screens with a quick menu
+ / -	Adjust the active screen's function	Zoom the chart, scroll descriptions, and more; see individual procedures
Hold  and press + / -	Increase / decrease LCD brightness	Settings(설정) → Key Bright(키 밝기) controls keypad lighting separately
Hold  and press 0	Save the current LCD screen	Useful when reporting a problem

3.2 Selecting and saving

Single-choice menus: Choose a value with $\uparrow / \downarrow$, then press $\rightarrow$. Check the selected-value indicator. Settings apply when selected; some menus return to the previous screen or restart MFNavis afterward. Pressing $\leftarrow$ does not restore the previous value.

Multiple-choice menus: In filter menus Catalogs(천체 목록) and Type(종류), each $\rightarrow$ press toggles the selected entry. Choose all the entries you need, then leave with $\leftarrow$. Select All(전체 선택) and Select None(선택 해제) select or clear all entries in the current list.

Commands with confirmation: Select Confirm(확인) and press $\rightarrow$ to execute, or select Cancel(취소) to return. Some commands execute without a confirmation screen. In INDI INIT(초기화), selecting a command and pressing $\rightarrow$ sends it immediately.

3.3 Quick menus and help

Hold $\square$ to open the quick menu for the current screen. Press the direction shown beside the function you want. When a quick submenu is open, a short $\square$ press closes one level; holding it closes the entire quick menu.



MFNavis quick menu · choose a function with its direction key

On screens with help, select **HELP** at the top of the quick menu. Use $\uparrow / \downarrow$ to change help pages and $\leftarrow$, $\rightarrow$, or $\square$ to close help. Help availability depends on the screen.

3.4 Check the active screen before using number keys

Number keys have different roles on different screens: object-number search in catalogs, numeric entry in forms, and movement in mount-control screens.

Current screen	Meaning of 0	Meaning of $\square$
Objects(천체) $\rightarrow$ selected object · object details	Stop mount movement, tracking, and Goto/Guide(GoTo/Guide) corrections	Cycle object views
Start(시작) $\rightarrow$ INDI $\rightarrow$ Guide(가이드)	Toggle Guide Correction(가이드 보정)	Sync the mount to the current pointing direction
Start(시작) $\rightarrow$ Align(정렬), during star selection	Cancel selection	Request alignment with the selected star
Start(시작) $\rightarrow$ Align (Day)(주간정렬)	Cancel / exit without saving	Start or save alignment

Current screen	Meaning of 0	Meaning of □
Settings(설정) → INDI Setting(INDI 설정) → Multi Align(멀티 정렬), during adjustment	Cancel / exit multi-point alignment	Confirm the current alignment point
SQM → quick menu CALIB → SQM Calibration(SQM 보정)	Cancel or skip sky frames, depending on the step	Advance / finish

In Guide(가이드), 0 does not stop all mount activity. Releasing a manual direction key stops that movement. In object details, 0 also stops tracking.

4 Preparing to observe with Start(시작)

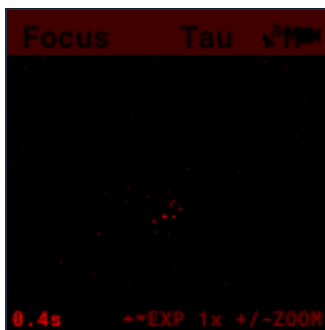
MNavis / Start



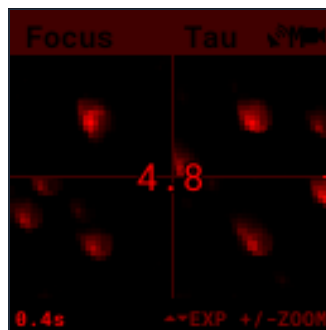
* Visible only when Mount Control is On.

4.1 Focus(초점): camera focus

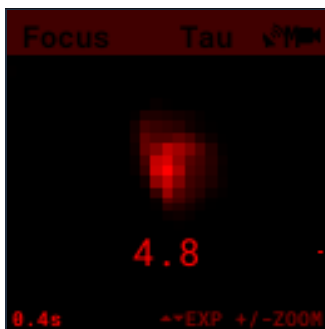
Path: Start(시작) → Focus(초점) · **Before you start:** Remove the lens cap and point the camera toward a star field.



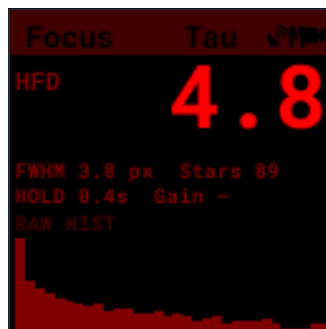
Focus(초점) · Image view



Focus(초점) · Stars view



Focus(초점) · Single view



Focus(초점) · Stats and HFD

1. Short-press to cycle through Image, Stars, Single, and Stats views. Use Stars to examine several stars and Single to enlarge one star.
2. If stars are hard to see, press to increase exposure. If the image is too bright or stars spread out, press to reduce exposure.
3. In Image, Stars, and Single views, use to change zoom.

4. Adjust the camera lens in small steps. With the same star, find the position where HFD decreases and the star image becomes compact. HFD measures how widely the star's light is spread.
5. Leave with . Exposure stays fixed while in Focus(초점); leaving restores the exposure setting used before entry.

To change gain, select → right Gain(이득) in the quick menu. Adjust exposure directly with / in Focus(초점).

Check the result: Stars should look small and sharp, with a stable HFD. Focus the telescope eyepiece separately.

4.2 Align(정렬): align the observing center at night

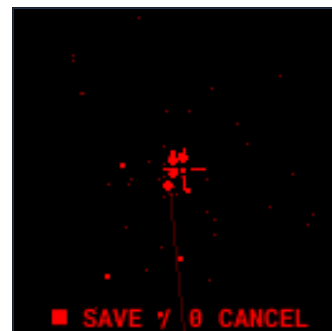
Path: Start(시작) → Align(정렬) · **Before you start:** Plate solving must be working. Center a bright star in the telescope eyepiece.



Align(정렬) · before pressing



Align(정렬) · choose a star with direction keys



Align(정렬) · star selected: save / 0 cancel

1. Press to start star selection.
2. Use the direction keys to select the same star that is centered in the eyepiece. Use / to zoom the chart.
3. Press again to request alignment.
4. Look for Aligned!(정렬 완료!) or Alignment requested. A request message alone does not guarantee completion; check the pointing display afterward.
5. After star-selection mode ends, leave with .

During star selection, selects a star to the left. Press to cancel selection. Press to reset the alignment point to the camera center.

Check the result: Center another bright object in the eyepiece and check that the guidance center matches your actual observing center. This procedure aligns camera and eyepiece centers. INDI mount multi-point alignment is covered in chapter 9.

4.3 Align (Day)(주간정렬): align the observing center in daylight

Path: Start(시작) → Align (Day)(주간정렬) · **Before you start:** Center a distant, easily recognized terrestrial target in the eyepiece.



Align (Day)(주간정렬) · 7 / 9 / 1 / 3 quadrants



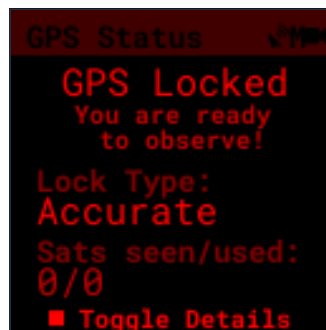
Align (Day)(주간정렬) · fine adjustment with direction keys

1. Press ☐ to begin.
2. Choose the quadrant containing the target using number keys: 7 upper left, 9 upper right, 1 lower left, 3 lower right. Quadrant selection can run up to three times.
3. Press a direction key once to enter fine adjustment. This first press does not move the reference point. Further direction-key presses move it onto the target.
4. Press ☐ to save and return to the previous menu.

Here, $+/-$ increase / decrease **exposure**. Pressing 0 cancels without saving a new alignment point. The quick menu can reset the reference point to the center or change exposure to Auto(자동).

4.4 GPS Status(GPS 상태)

Path: Start(시작) → GPS Status(GPS 상태) or Tools(도구) → Place & Time(위치/시간) → GPS Status(GPS 상태)



GPS Status(GPS 상태) · receiver information

Check the location-fix status and satellite information. If a fix is unavailable, wait with an open view of the sky. Where GPS is unavailable, enter location and time manually as described in chapter 11. Leave with ☐.

4.5 INDI

Path: Start(시작) → INDI · **Visibility:** Tools(도구) → Experimental(실험적) → Mount Control(가대 제어) → On(켜짐)

The menu contains STATUS(상태), INIT(초기화), and Guide(가이드). Changing Mount Control(가대 제어) restarts MFnavis. Connection, automatic movement, mount alignment, and Guide(가이드) controls are described in chapter 9.

5 Viewing the sky with Chart(성도)

Path: Chart(성도) · **Before you start:** A pointing direction must be available to draw the chart.



Chart(성도) · stars and chart at the current pointing direction

Control	Action
+	Zoom in
-	Zoom out
□	Restore the default field of view
→	Open details of the object at the chart center
←	Return to the previous menu
Hold □	Open the chart quick menu

As you move the telescope, the chart updates to the current direction. Change orientation, reticle, constellation lines, deep-sky objects, and coordinate display in Settings(설정) → Chart...(성도...) .

Opening details with → requires **Center Object(중앙 천체) to be On(켜짐)** and a selectable object at the chart center. If No solve(해 없음) appears, check Focus(초점) and plate-solving status first. Holding → opens the most recently viewed object instead of the object at the chart center.

6 Finding targets with Objects(천체)

MFNavis / Objects



Object list → RIGHT → Object details → RIGHT → LOG

6.1 Choosing an object list

Menu	How to use it
All Filtered(필터 결과)	Open all objects matching the current filters. Select with ↑ / ↓ and open details with →.
By Catalog(카탈로그별)	Select a catalog and press →. Includes Planets(행성), Comets(혜성), NGC, Messier, DSO..., and Stars...(별...).
Recent(최근)	Reopen objects viewed during the current run. Empty if no objects have been viewed.
Obs Lists(관측목록)	Select a previously supplied observing-list file and open it with →, then select an object.
Custom(사용자)	Enter RA and Dec for a target outside the catalogs. Move between fields with ↑ / ↓, enter numbers, and confirm with →.
Name Search(이름 검색)	Enter a name and press → to open results, then select an object.
Set Filters(필터 설정)	Choose catalogs, object types, altitude, magnitude, and observation status.



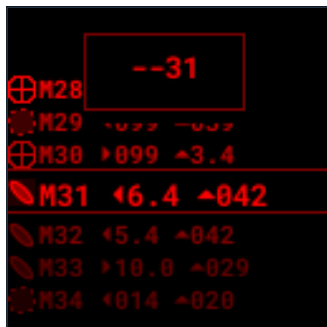
Objects(천체) · lists and search menus

Catalog groups

Group	Included catalogs
Direct entries	Planets(행성), Comets(혜성), NGC, Messier
DSO...	Abell Pn, Arp Galaxies(Arp 은하), Barnard, Caldwell, Collinder, E.G. Globbs(E.G. 구상), Harris Globbs(Harris 구상), Herschel 400, IC, Lynga Opn Cl(Lynga 산개), Messier, NGC, Sharpless, TAAS 200
Stars...(별...)	Bright Named(밝은 별), SAC Doubles(SAC 이중성), SAC Asterisms(SAC 성군), SAC Red Stars(SAC 적색성), RASC Doubles(RASC 이중성), WDS Doubles(WDS 이중성), TLK 90 Variables(TLK 변광성)

6.2 Finding objects by number and changing the list view

In a catalog list, number keys jump to an object near that number. For example, press **3**, then **1** in Messier, **check that the selected name is M31**, and press **→**. If filters hide the target, that number may select a different object.



Messier · entering M31

Press **□** to change the list view. While a numeric-entry indicator is visible, **□** clears it. Select **hold □ → left** Sort(정렬) to choose Nearest(가까운순) or Standard(표준) sorting; right Filter(필터) opens filters directly. The Comets(혜성) quick menu also offers Refresh(새로고침).

6.3 Name Search(이름 검색): search by name

Path: Objects(천체) → Name Search(이름 검색)



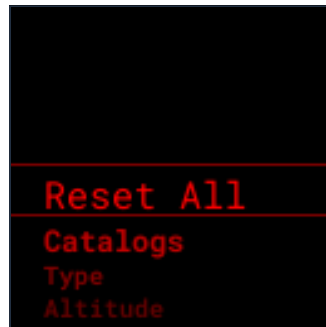
Name Search(이름 검색) ·
character-key layout

1. Enter a name with the number keys using the displayed letter layout. In Multi-Tap, press the same key repeatedly to select a letter. In T9, press the corresponding key once for each letter.
2. Press **-** to delete the last character and **+** to insert a space. Press **□** to change the character layout.
3. Press **→** to open results. Select a result with **↑ / ↓** and open details with **→**.
4. Return from results to the search-entry screen to edit the name and search again.

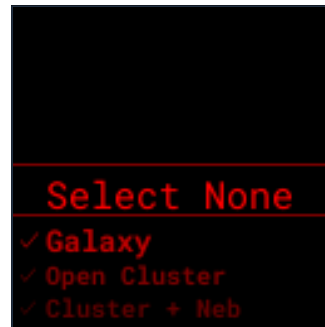
Choose the input method in **Settings(설정) → User Pref...(사용자...) → Search Input(입력 방식)**. You can also enter text with an external keyboard.

6.4 Set Filters(필터 설정): filter lists

Path: **Objects(천체) → Set Filters(필터 설정)**



Set Filters(필터 설정) · filter
entries



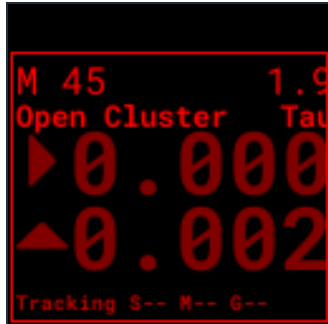
Type(종류) · multiple selections

Entry	Control and effect
Reset All(전체 초기화)	Select Confirm(확인) and press → to restore default filters; Cancel(취소) returns.
Catalogs(천체 목록)	Use → to select / clear multiple catalogs included in All Filtered(필터 결과).
Type(종류)	Use → to select / clear object types such as galaxies, clusters, nebulae, stars, planets, and comets.
Altitude(고도)	Choose a minimum altitude: None(없음) or 0°, 10°, 20°, 30°, 40°.
Magnitude(등급)	Choose the maximum magnitude shown: None(없음) or 6–15. Larger numbers include fainter objects.
Observed(관측됨)	Choose Any(전체), Observed(관측됨), or Not Observed(미관측) to filter by observation records.

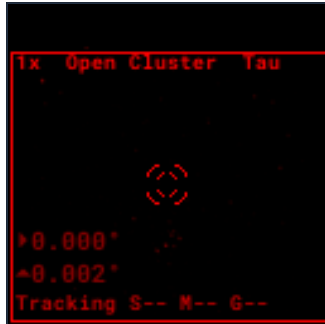
Name Search(이름 검색) and **Recent(최근)** are unaffected by the ordinary list filters. Check filters first if an object is missing from a catalog or observing list. Altitude filtering requires the correct location and time.

6.5 Object details and movement guidance

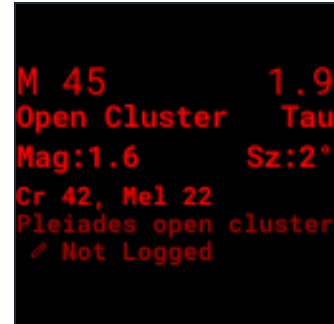
Path: Objects(천체) → an object list → select an object → →



M45 · movement guidance and tracking border



M45 · camera view and tracking border



M45 · description view

Each ☐ press cycles through **movement guidance** → **camera** → **object image** → **description** → **contrast information**. In guidance view, follow the direction arrows and remaining angle toward the target.

Control	Action
↑ / ↓	Previous / next object in the current list
+ / -	Zoom in / out in camera view; next / previous portion of a description; change eyepiece in other views
→	Open LOG(로그) when a current pointing direction is available
←	Return to the list
5	Request GoTo to the selected object using the connected mount
0	Stop mount movement, tracking, and Goto/Guide(GoTo/Guide) corrections
7	Request mount Sync using the current pointing direction
1	Cycle GoTo Type(자동 도입 유형)
8 / 2 / 4 / 6	Move the mount north / south / west / east while held
9 / 3	Increase / decrease manual mount movement speed

Mount controls require Mount Control(가대 제어) to be enabled and a connected mount. When GoTo Type(자동 도입 유형) is Off(꺼짐), 5 does not initiate movement. After a request message, also check actual movement and arrival status.

Tracking border in the Push view: After GoTo completes, a rectangular border appears below the title bar around the guidance and camera views when the mount is tracking the currently selected object.

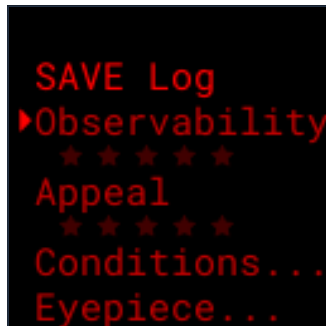
- **Appearance:** A dark outer line and bright inner line remain visible against bright camera images and dark backgrounds. The bright line follows the display color setting and remains visible in night mode.

- **When it disappears:** Tracking stops, a new GoTo or manual movement begins, the mount parks, an error or disconnect occurs, or fresh status information is unavailable. Viewing a different object from the tracked target also hides the border.
- **Checking arrival:** The border is absent during ordinary GoTo motion and pulse-guide arrival corrections. Once movement completes, check the border, remaining angle, and status at the bottom together.

To align the eyepiece center using the selected object, center it in the eyepiece and select **hold** ☐ **→ down ALIGN(정렬)** **→ right ALIGN(정렬)**. Left **CANCEL(취소)** in the submenu returns. This **ALIGN(정렬)** aligns camera and eyepiece centers; mount Sync is a separate command.

6.6 LOG(로그): record an observation

Open LOG(로그) with **→** in object details, then select a field with **↑ / ↓**. For ratings, enter **0–5** or cycle values with **→**. Press **→** on observing conditions or eyepiece to open a selection screen. Select the save entry and press **→**; Logged!(기록됨!) appears and the details screen returns.



LOG(로그) · observation ratings

Conditions(조건) includes Transparency(투명도) and Seeing(시상). Choose NA(해당없음), Excellent(최상), Very Good(매우 좋음), Good(좋음), Fair(보통), or Poor(나쁨) to describe the conditions, then apply with **→**.

6.7 Custom(사용자): enter coordinates

Path: **Objects(천체) → Custom(사용자)**



Custom(사용자) · RA / Dec fields

Move between fields with **↑ / ↓** and enter RA and Dec with the number keys. Press **-** to delete a digit. In the Dec degrees field, **+** changes the sign; in Epoch, **+** changes the coordinate reference epoch. Press **□** to change the coordinate-entry format. Check every value and confirm with **→**, or leave without saving with **←**.

7 Checking sky brightness with SQM

Path: `SQM` · **Meaning:** Shows the brightness of the sky toward which the camera points, in mag/arcsec². A larger number means a darker sky.



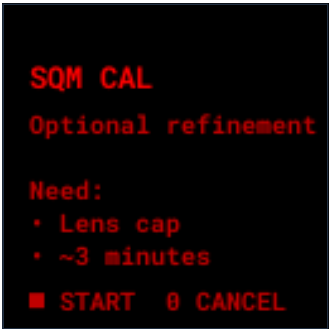
SQM · sky-brightness value and units

Control	Action
<code>□</code>	Switch between measurement and sky-condition description
<code>+ / -</code>	Next / previous portion of the description
<code>←</code>	Return to the previous menu
Hold <code>□</code> → left CALIB	Open the SQM calibration wizard
Hold <code>□</code> → down SWEEP	Open exposure-based SQM diagnostic measurements

Moonlight, clouds, twilight, and the camera’s pointing altitude affect readings. Compare changes at one location using similar directions and conditions. Basic brightness measurement may continue even if plate solving fails. Entering SQM switches exposure behavior for measurement; leaving restores normal exposure behavior.

7.1 SQM Calibration(SQM 보정)

Path: `SQM` → hold `□` → left CALIB → SQM Calibration(SQM 보정)



SQM Calibration(SQM 보정) · introduction

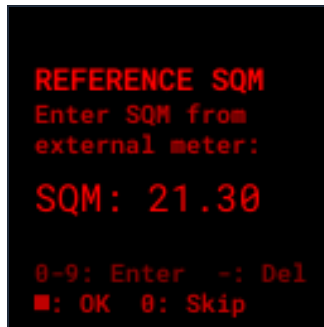
1. Open CALIB and press `□` to start.

2. At the lens-cap instruction, put the cap on and press . Wait for frame collection to finish.
3. At the cap-removal instruction, remove the cap and press to collect sky frames.
4. Review the analysis and results, then leave with .

cancels at the introduction and cap-on instruction. At the cap-off instruction or during sky-frame collection, it **skips sky frames and proceeds to analysis**. Do not use as a cancel button during cap-on frame collection.

7.2 SQM Sweep: diagnostic measurements

Path: SQM → hold → down SWEEP → SQM Sweep



SQM Sweep · enter 2130 for a
21.30 reference

If you know a reference SQM value, enter four digits: for example, means **21.30**. Press to delete the last digit and to confirm. With an empty entry, or continues without a reference value.

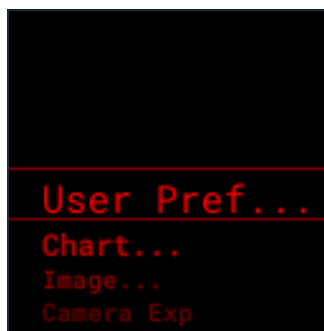
At confirmation, press to start collection; press again after completion to leave. Here, cancels at confirmation. SWEEP helps diagnose how readings change with exposure.

8 Changing display and observing settings

MFNavis / Settings



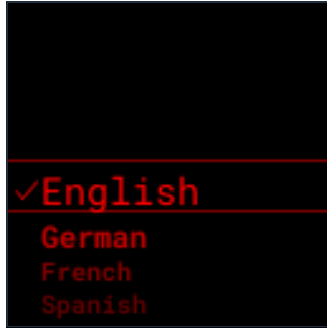
* Visible only when Mount Control is On. Values are listed in the manual.



Settings(설정) · settings menu

8.1 User Pref...(사용자...): preferences

Path: Settings(설정) → User Pref...(사용자...) → entry → choose a value → →



Language(언어) · selected-value indicator

Entry	Values	Purpose
Key Bright(키 밝기)	-4-3	Keypad lighting. For LCD brightness, hold  and press + / -
Sleep Time(절전 시간)	Off(꺼짐), 10s, 20s, 30s, 1m, 2m	Idle time before sleep
Menu Anim(메뉴 효과)	Off(꺼짐), Fast(빠름), Medium(보통), Slow(느림)	Menu transition speed
Scroll Speed(스크롤 속도)	Off(꺼짐), Fast(빠름), Medium(보통), Slow(느림)	Scrolling speed for long text
Search Input(입력 방식)	Multi-Tap, T9	Text-entry method for name search
Az Arrows(방위 화살)	Default(기본), Reverse(반전)	Azimuth arrow direction in movement guidance
Language(언어)	English(영어), German(Deutsch), French(Français), Spanish(Español), Korean(한국어), Chinese(中文)	LCD UI language; select English(영어) for English

8.2 Chart...(성도...): chart display

Path: Settings(설정) → Chart...(성도...) → entry → choose a value → →

Entry	Values and effect
Coordinate Sys.(좌표계)	Horizontal(지평좌표): horizon orientation; EQ (Auto)(적도(자동)): automatic equatorial orientation; EQ (North-up)(적도(북위)) / EQ (South-up)(적도(남위)): north / south celestial pole upward
Reticle(십자선)	Off(꺼짐) / Low(낮음) / Medium(보통) / High(높음): central reticle brightness
Constellation(별자리)	Off(꺼짐) / Low(낮음) / Medium(보통) / High(높음): constellation-line brightness
DSO Display(DSO 표시)	Off(꺼짐) / Low(낮음) / Medium(보통) / High(높음): deep-sky-object display brightness
RA/DEC Disp.(RA/DEC 표시)	Off(꺼짐) / HH:MM / Degrees(도): coordinate display format
Center Object(중앙 천체)	Off(꺼짐) / On(켜짐): central-object name and opening details with → from Chart(성도)

Before location is available, chart orientations that require it may use a temporary orientation. Check the upward direction again after acquiring a GPS location.

8.3 Image...(이미지...): object images

In Settings(설정) → Image...(이미지...) , enable **NSEW Labels(방위 표시)** with On(켜짐) for direction labels, and **Object Size(천체 크기)** with On(켜짐) for object-size and orientation outlines. Choose Off(꺼짐) to disable them. Open each entry and select the desired value with → .

8.4 Camera Exp(노출) and Camera Gain(Gain)

Path	Values	Control
Settings(설정) → Camera Exp(노출)	Auto(자동), Star(항성), 0.025s, 0.05s, 0.1s, 0.2s, 0.4s, 0.8s, 1s	Select with ↑ / ↓, apply with → . Auto(자동) and Star(항성) are automatic exposure methods; numeric values are fixed durations
Settings(설정) → Camera Gain(Gain)	Profile(설정 묶음), 1x, 2x, 4x, 8x, 12x, 15x, 16x, 20x, 22x, 24x, 30x	Select with ↑ / ↓, apply with → . Profile(설정 묶음) uses the camera profile's reference value

Longer exposures can capture more stars but may elongate them during movement. Actual supported exposure and gain ranges depend on the sensor. Temporary exposure in Focus(초점) is separate from the observing exposure setting in Settings(설정).

8.5 WiFi Mode(WiFi 모드) and Mount Type(가대 종류)

Entry	How to use it	Check afterward
WiFi Mode(WiFi 모드) → Client Mode(Client 모드)	Connect to an existing wireless network	Check the address in Tools(도구) → Status(상태)
WiFi Mode(WiFi 모드) → AP Mode(AP 모드)	Let MFNavis provide a wireless access point	Connect to MFNavisAP, then open http://10.10.10.1
WiFi Mode(WiFi 모드) → AP+STA Mode(AP+STA 모드)	Use an access point and an existing network together	Check both connections in Status(상태)
Mount Type(가대 종류) → Alt/Az(경위대)	Select an altitude-azimuth telescope mount	Check the value after MFNavis restarts
Mount Type(가대 종류) → Equatorial(적도의)	Select an equatorial telescope mount	Check the value after MFNavis restarts

Changing WiFi mode may disconnect your phone or computer. Reconnect to the network and address appropriate for the new mode.

9 Connecting and moving an INDI mount

INDI MOUNT detailed settings: Open the MFNavis web server on a phone or PC and select **INDI**. You can configure USB or network connections, mount limits, multi-point alignment, backlash, GoTo / Guide, and SkySafari. See 14.1 for web access, 14.4 for connections, and 14.5–14.7 for the other settings.

9.1 Enabling mount control

1. Select **Tools(도구) → Experimental(실험적) → Mount Control(가대 제어) → On(켜짐)** . MFNavis restarts.
2. Request connection with **Start(시작) → INDI → INIT(초기화) → Connect(연결)** .
3. In **Start(시작) → INDI → STATUS(상태)** , check connection, coordinates, tracking, and Home / Park(파크) status.
4. Once location and time are ready, select **INIT(초기화) → Set Location(위치 설정)** to send them to the mount.
5. To use a parked mount, select **INIT(초기화) → Unpark(언파크)** and check status.



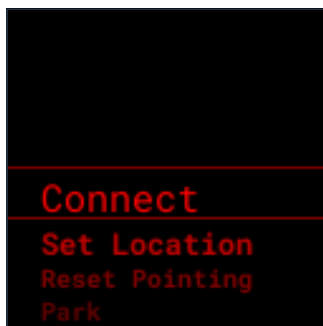
```
State    connected
Msg      Ready
Age      0s ago
Device   Example mount
Home     Home
Park     Unparked
Raw      --
RA       56.75
Dec      24.12
```

STATUS(상태) · connection status

When Mount Control(가대 제어) is Off(꺼짐), **Start(시작) → INDI** and **Settings(설정) → INDI Setting(INDI 설정)** are hidden. For a first connection or a changed transport, configure the web **INDI → LX200 OnStepX Driver Connection(LX200 OnStepX 드라이버 연결)** section first (14.4). Its heading also includes the current driver's name. Check STATUS(상태) for successful connection even after an LCD request message appears.

9.2 INIT(초기화) commands

Path: **Start(시작) → INDI → INIT(초기화) → select a command → →**



```
Connect
Set Location
Reset Pointing
Park
```

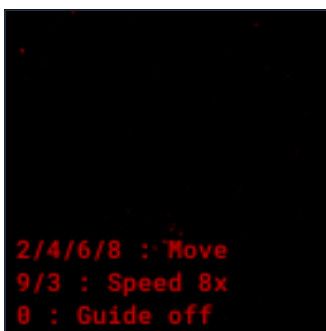
INIT(초기화) · command selection

Entry	Purpose	Check afterward
Connect(연결)	Request INDI mount connection and initialization	Connection and coordinates in STATUS(상태)
Set Location(위치 설정)	Send current location and time to the mount	Synchronization status and error notices
Reset Pointing(좌표 초기화)	Request rebuilding of the current pointing-coordinate reference	Updated pointing display
Park(파크)	Request movement to the configured park position	Park(파크) status in STATUS(상태)
Unpark(언파크)	Request unparking	Unpark(언파크) status in STATUS(상태)
Set Home(홈 설정)	Request setting the current position as Home	Home status and device response
Return Home(홈으로 복귀)	Request movement to Home	Completed movement and Home status
Set-Park(파크 위치 설정)	Request setting the current position as the park position	Device response and park settings
Restart INDI(INDI 재시작)	Request an INDI driver restart	Reconnection and received coordinates

Support varies by mount and driver. Park(파크) and Return Home(홈으로 복귀) can move the telescope; check the movement path before executing them.

9.3 Guide(가이드): manual movement and Sync

Path: Start(시작) → INDI → Guide(가이드)



Guide(가이드) · movement / speed / correction hints

MFNavis / INDI Guide



8 / 2 / 4 / 6: hold to move N / S / W / E

Release direction key: stop manual motion

9 / 3: speed up / down

SQUARE: Sync 0: Guide Correction toggle

Object details: 5 = GoTo, 0 = Stop + tracking off

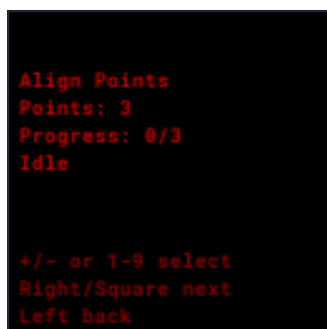
Control	Action in Guide(가이드)
Hold 8 / 2 / 4 / 6	Move north / south / west / east; release to stop manual movement
9 / 3	Increase / decrease manual movement speed
□	Sync the mount to the currently available sky pointing direction
0	Toggle Guide Correction(가이드 보정)
←	Return to the previous menu; leaving stops manual movement

An external letter keyboard can use `q / w / e` , `a / s / d` , and `z / x / c` as a direction pad. `s` stops manual movement; `,` / `.` decrease / increase speed. Custom key mappings can change these actions.

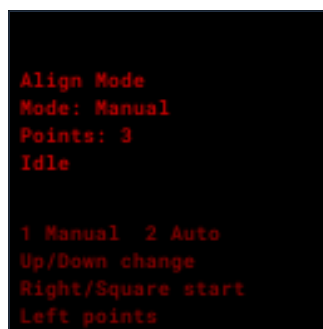
If a current direction is unavailable for Sync, No solve(해 없음) appears. In Guide(가이드), □ sends Sync rather than switching views.

9.4 Multi Align(멀티 정렬): multi-point alignment

Path: Settings(설정) → INDI Setting(INDI 설정) → Multi Align(멀티 정렬)



Multi Align(멀티 정렬) · point count



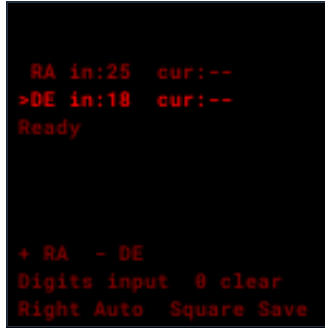
Multi Align(멀티 정렬) · Manual(수동) / Auto(자동)

1. Adjust the number of alignment points with number keys or `+ / -` . Press `→` or `□` to continue.
2. Select Manual(수동) / Auto(자동) with `↑ / ↓` , then begin with `→` or `□` . At this step, `1` also starts Manual(수동), and `2` starts Auto(자동).
3. In Manual(수동), choose a star with `↑ / ↓` and move to it with `→` or `□` . In Auto(자동), follow the instructions during preparation and movement.
4. On the adjustment screen, use mount direction controls to center the star. Adjust speed with `9 / 3` and **confirm the current point with □**.
5. Complete the required points and check completion status.

During adjustment, `←` returns to star selection in manual mode; in automatic mode, it cancels the current process and returns to mode selection. At the adjustment step, `0` cancels alignment and exits. Star lists and automatic startup depend on location, time, plate solving, available stars, and mount status.

9.5 Backlash(백래시)

Path: Settings(설정) → INDI Setting(INDI 설정) → Backlash(백래시)



Backlash(백래시) · RA / DE entry

Select the RA axis with **+** or DE axis with **-**. Enter values with number keys and press **→** to send both axes. **→** requests automatic backlash measurement for the selected axis. The input range is 0–999; **0** clears the selected axis's input. **Here, 0 is not entered as a digit.** Check the device's values, units, and support for automatic measurement.

9.6 Goto/Guide(GoTo/Guide) settings

Path: Settings(설정) → INDI Setting(INDI 설정) → Goto/Guide(GoTo/Guide) → entry → choose a value → →

Entry	Values	Meaning
GoTo Type(자동 도입 유형)	Off(꺼짐) / INDI Mount(INDI 가대) / MFNavis	Disable automatic movement / use mount GoTo / use MFNavis coordinate-based movement and corrections
Tracking Guide(추적 가이드)	Off(꺼짐) / On(켜짐)	Correct the target position during tracking
GoTo Recovery(GoTo 복구)	Off(꺼짐) / On(켜짐)	Use GoTo to return after a large target deviation
Recovery Range	0.25°, 0.5°, 1°, 2°, 3°	Deviation angle used to decide GoTo recovery
Manual Re-target(수동 재타겟)	Off(꺼짐) / On(켜짐)	Update the tracking target to a new direction after manual movement
Max GoTos(최대 GoTo 횟수)	3, 5, 10, 15, 20	Maximum repeated moves during MFNavis GoTo
Invert Guide RA/Az(가이드 RA/Az 반전)	Off(꺼짐) / On(켜짐)	Reverse RA / azimuth correction direction
Invert Guide Dec/Alt(가이드 Dec/Alt 반전)	Off(꺼짐) / On(켜짐)	Reverse Dec / altitude correction direction

After reversing a correction direction, check with small movements that target error decreases. Manual Re-target(수동 재타겟) changes the tracking target; it does not automatically change the catalog object selected in object details.

10 Configuring hardware in Settings(설정)

10.1 Advanced(고급): hardware and input devices

Path: Settings(설정) → Advanced(고급)



Advanced(고급) · hardware and input-device menus

Entry	Control and checks
MFNavis Type(MFNavis형)	Choose the actual assembly orientation: Left(왼쪽), Right(오른쪽), Straight(직선형), Flat v3, Flat v2, or AS Bloom, then press →. Check orientation after restart
Camera Type(카메라 종류)	Choose the installed sensor and press →: IMX678 (Auto(자동)), v2 - imx477, v3 - imx296 Mono(v3 - imx296 모노), v3 - imx296 Color(v3 - imx296 컬러), v3 - imx462 Mono(v3 - imx462 모노), v3 - imx462 Color(v3 - imx462 컬러). Sensor changes can require reboot; some variant changes restart the software. Follow the screen instructions
Lens	Use the lens selection and measurement procedure below
Distortion	Use the distortion calibration procedure below
GPS Settings(GPS 설정)	Match GPS Type(GPS 종류), GPS Baud Rate(GPS Baud), and GPS Port(GPS 포트) to the receiver
Time Sync(시간 동기화)	Configure time synchronization and its sources
Bluetooth(블루투스)	Scan, pair, and reconnect external input devices
Joystick(조이스틱)	Test buttons and assign them to functions
Keyboard	Test keys and assign them to functions
WiFi Recover(WiFi 복구)	Request recovery with Confirm(확인) →; return with Cancel(취소)

Camera Mono / Color refers to the sensor's monochrome or color variant, rather than a display color option.

10.2 Lens: selection and measurement

Path: Settings(설정) → Advanced(고급) → Lens

Choice	How to use it
4mm, 6mm, 8mm, 10mm, 12mm, 16mm, 25mm	Choose the camera lens's focal length and press →
Manual(수동) (mm)	Enter a focal length and confirm with ←. Delete the last character with -; switch the character layout with □ to enter digits and a decimal point
Auto(자동) (Measure)	Start measurement with →. Point toward stars in a sky field that can be plate-solved and check progress

Automatic measurement collects usable frames and evaluates the result. Check the measured focal length afterward. During measurement, ←, □, or 0 cancels and exits. **Lens values describe the MFNavis camera lens**; manage the telescope focal length in Equipment(관측 장비).

10.3 Distortion: calibration

Path: Settings(설정) → Advanced(고급) → Distortion

1. Select Status(상태) to check calibration for the current camera and lens.
2. Point at a star field, select Measure Sky, and press → to measure.
3. Check collection progress and results. After completion, check Status(상태) to confirm application.
4. During measurement, ←, □, or 0 cancels and exits. Cancel(취소) Measurement in the menu also requests cancellation.

Reset clears the calibration when Confirm(확인) is selected; Cancel(취소) returns. Check the calibration for your current combination after changing the camera or lens.

10.4 GPS Settings(GPS 설정) and Time Sync(시간 동기화)

Path	Values and control
GPS Settings(GPS 설정) → GPS Type(GPS 종류)	Choose UBlox / GPSD (generic)(GPSD), then →. MFNavis restarts
GPS Settings(GPS 설정) → GPS Baud Rate(GPS Baud)	Choose 9600 (standard) / 115200 (UBlox-10) to match your receiver, then →
GPS Settings(GPS 설정) → GPS Port(GPS 포트)	Choose Auto(자동) or the connected port, then →. Ports: ttyAMA1, ttyAMA2, ttyAMA3, serial0, ttyAMA0, ttyAMA10, ttyS0, ttyACM0, ttyUSB0
Time Sync(시간 동기화) → Time Sync(시간 동기화)	Select Off(꺼짐) / On(켜짐) for time synchronization
Time Sync(시간 동기화) → Chrony Source(Chrony 소스)	Select Off(꺼짐) / On(켜짐) for the Chrony time source
Time Sync(시간 동기화) → GPS Source(GPS 소스)	Select Off(꺼짐) / On(켜짐) for the GPS time source

Path	Values and control
Time Sync(시간 동기화) → RTC Sync(RTC 동기화)	Select Off(꺼짐) / On(켜짐) for RTC synchronization

After applying changes, check reception and synchronization in **Tools(도구) → Place & Time(위치/시간) → GPS Status(GPS 상태) / Time Sync(시간 동기화)**. An incorrect port or baud rate prevents GPS reception.

10.5 Bluetooth(블루투스): connect input devices

1. Open **Settings(설정) → Advanced(고급) → Bluetooth(블루투스)** and put the external keyboard into pairing mode.
2. Select Scan and press **→**. Choose the device and press **→** to open its action menu.
3. Select Pair+Connect(페어+연결), press **→**, and follow the pairing instructions.
4. Check connection status and test actual key presses.

Reconnect(재접속) reconnects a known device; Refresh(새로고침) updates the list. In a device's action menu, Connect(연결) / Disconnect(연결 해제) connect / disconnect, Pair Again(다시 페어) repeats pairing, and Remove(제거) removes registration. Press **←** to close the action menu or pairing.

10.6 Joystick(조이스틱) and Keyboard mappings

Screen	How to use it
Joystick(조이스틱)	Open Test Buttons(버튼 확인) with → to test buttons. Select a function, press → , then press the desired button on the connected device to assign it.
Keyboard	Open Test Keys with → to test keys. Select a function, press → , then press the desired external key to assign it.

Press **←** to return from tests or assignment waiting. **Clear All(전체 지우기)** clears custom mappings for that device when you press **→**. After assignment, check the menu's updated label and test the action on its actual screen.

10.7 IMU Settings(IMU 설정): movement sensor

Path: **Settings(설정) → IMU Settings(IMU 설정)**

Entry	Control and effect
Sensitivity(감도)	Off(꺼짐) / Very Low(아주 낮음) / Low(낮음) / Medium(보통) / High(높음): movement-detection sensitivity; changes restart MFNavis
Compass(나침반)	Off(꺼짐) / On(켜짐): enable the sensor's compass; changes restart MFNavis
Calibration(캘리브레이션) → Save(저장)	Request saving the current sensor calibration
Calibration(캘리브레이션) → Load(불러오기)	Request loading saved sensor calibration

Entry	Control and effect
Calibration(캘리브레이션) → Clear(지우기)	Request deleting saved sensor calibration

Select each command and press  to run it. These functions require the corresponding physical sensor.

11 Checking status and managing MFNavis with Tools(도구)

MFNavis / Tools



11.1 Status(상태)

Path: Tools(도구) → Status(상태)

```
LAST SLV      0.0s C 12
RA/DEC        03h47m/24.12
AZ/ALT        120.00/ 45.00
WIFI          AP+STA
IP            NONE
SSID          UNKN
IMU           --
IMU AGE       --
IMU CAL       --
```

Status(상태) · device-status
summary

Check current plate-solving, location, communications, and equipment status. After changing WiFi mode, check the connection address. When troubleshooting, check both Status(상태) and the relevant function's status screen. Return with ←.

11.2 Equipment(관측 장비): telescope and eyepiece

Path: Tools(도구) → Equipment(관측 장비)



Equipment(관측 장비) · telescope and eyepiece

Choose the telescope or eyepiece row with $\uparrow / \downarrow$, then open its selection list with $\rightarrow$. Select the equipment and press $\rightarrow$, then return to Equipment(관측 장비) to check magnification and field of view. Lists depend on your saved equipment configuration. Register and edit equipment in the web equipment-management screen.

11.3 Place & Time(위치/시간): location and time

Path: Tools(도구) $\rightarrow$ Place & Time(위치/시간)



Enter Coords(좌표 입력) · latitude entry



Set Time/Date(시간/날짜) · time entry

Entry	How to use it
GPS Status(GPS 상태)	Check location-fix status and satellite information; return with $\leftarrow$.
Time Sync(시간 동기화)	Check time synchronization status. Configure it in Settings(설정) $\rightarrow$ Advanced(고급) $\rightarrow$ Time Sync(시간 동기화).
Set Location(위치 설정) $\rightarrow$ Enter Coords(좌표 입력)	Enter latitude $\rightarrow$ longitude $\rightarrow$ altitude. Enter digits and press $\rightarrow$ to advance fields and screens. Confirming the final altitude applies the location.
Set Location(위치 설정) $\rightarrow$ Load Location(위치 불러오기)	Select a saved place with $\uparrow / \downarrow$ and open its action menu with $\rightarrow$. Select Load(불러오기) and press $\rightarrow$. The loaded place also becomes the default.
Set Location(위치 설정) $\rightarrow$ Save Location(위치 저장)	Enter a name for the current location and confirm with $\leftarrow$.
Set Time/Date(시간/날짜)	Acquire a location first, enter local time, then press $\rightarrow$ for the date. Both take effect after confirming the date.
Reset Location(위치 초기화)	Clear the current location, then acquire it again through GPS or manual entry.

Entry	How to use it
Reset Time/Date(시간 초기화)	Clear current time/date status, then set it again from a time source or manual entry.

In coordinate entry, use **+** to change sign and check N / S and E / W indicators. **-** deletes the last digit in the current field. In numeric entry, **←** moves to the previous field or cancels at the first field. **Time entry cannot proceed until a location is available.** Use local time at the observing location.

A saved place's action menu also offers Rename(이름 변경) and Delete(삭제). Confirm the new name with **←**. Selecting Delete(삭제) and pressing **→** deletes the place directly; **←** closes the action menu.

11.4 Console and Software Upd(SW 업데이트)

Console: Open **Tools(도구) → Console** to read device messages. Use **↑ / ↓** for older / recent messages and **←** to return. Number keys run test actions and can change time status; use direction keys when reading messages during real observations.

Software Upd(SW 업데이트): With internet access, open **Tools(도구) → Software Upd(SW 업데이트)**. If an update is offered, select update or Cancel(취소) with **↑ / ↓** and press **→**. Updating cannot start if no update is available or release information cannot be fetched. If an OS migration confirmation appears, review its conditions before continuing. Maintain power throughout installation.

11.5 Test Mode(테스트 모드)

Path: **Tools(도구) → Test Mode(테스트 모드)**

Use stored images for indoor demonstrations or checks. Select the entry and press **→**. This mode uses test images instead of the actual sky; check that it is disabled before observing.

11.6 Experimental(실험적)

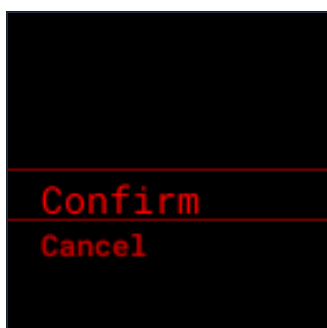
Entry	How to use it
Polar Align(극축정렬)	Assist equatorial polar alignment using the procedure below
Mount Control(가대 제어)	Select Off(꺼짐) / On(켜짐) with → . INDI menu visibility changes after restart.
Dev Tools(개발도구) → Telemetry → Record(기록)	Off(꺼짐) / On(켜짐) disables / enables coordinate and sensor recording.
Dev Tools(개발도구) → Telemetry → Images(이미지)	Off(꺼짐) / On(켜짐) controls image recording.
Dev Tools(개발도구) → Telemetry → Load(불러오기)	Select a saved recording and press → to replay. Select Stop replay(리플레이 중지) in the same list and press → to stop. Used for reproducing and diagnosing problems.

Polar Align(극축정렬) path: **Tools(도구) → Experimental(실험적) → Polar Align(극축정렬)**

Press to begin the instructions. Rotate the equatorial setup, stop at each position, and press to request a plate-solving measurement. After the required measurements, adjust the mount's polar-alignment controls using the displayed altitude and azimuth corrections. cancels the current collection while waiting for a measurement, or resets at other steps. At the AIM step with at least two collected points, can also calculate corrections. On the correction screen, starts a new measurement, so review the result first.

11.7 Power(전원): shutdown and restart

Task	Path and control
Shut down	Tools(도구) → Power(전원) → Shutdown(종료) → Confirm(확인) →
Restart	Tools(도구) → Power(전원) → Restart(재시작) → Confirm(확인) →
Return without executing	Cancel(취소) → on either confirmation screen



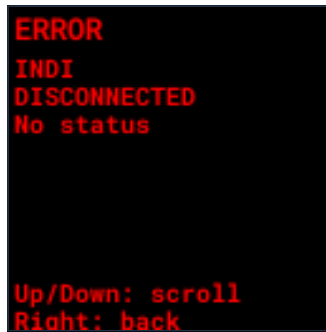
Shutdown(종료) · Confirm(확인) /
Cancel(취소)

Shutdown(종료) performs a normal shutdown; Restart(재시작) restarts the system. Wait for shutdown to complete before removing power.

12 Troubleshooting

12.1 Closing an error notice

LCD error notices remain until acknowledged. Read long notices with  / , then close with , , or . Closing returns to the previous screen; **it does not retry the failed command**. Investigate the cause, then request the action again if needed.



ERROR(오류) · reading and closing the notice

12.2 Checks by symptom

Symptom	Checks and action
Cannot return from a menu	Check for active entry or alignment and use its cancellation procedure. During ordinary navigation, hold  for the top-level menu.
No stars visible	Check lens cap and sky direction → exposure and focus in Focus(초점).
No solve(해 없음) appears	Check star images in Focus(초점) → camera and lens settings → Status(상태).
An object is missing	Search with Name Search(이름 검색) → check altitude, magnitude, and type in Set Filters(필터 설정) → use Reset All(전체 초기화) if needed.
→ does nothing in Chart(성도)	Check Center Object(중앙 천체) is On(켜짐), plate solving is working, and a central object exists.
INDI menu is missing	Check Tools(도구) → Experimental(실험적) → Mount Control(가대 제어) is On(켜짐).
GoTo does not start	Check target selection in object details → whether GoTo Type(자동 도입 유형) is Off(꺼짐) → connection and Park(파크) in INDI STATUS(상태) → error notices.
Pointing is incorrect after movement	Check camera/eyepiece alignment, location/time, and Mount Type(가대 종류); check INDI mount alignment if needed.
No GPS fix	Check open sky → GPS Type(GPS 종류), baud rate, and port → enter location/time manually if needed.
Time entry cannot proceed	Acquire an observing location first; confirm both local time and date.
Connection lost after WiFi change	Reconnect to the new mode's network → check the address in Status(상태) → use WiFi Recover(WiFi 복구) if needed.

Symptom	Checks and action
External keys behave unexpectedly	Check Keyboard / Joystick(조이스틱) tests and mappings; also check the active screen's number-key meanings.

When reporting a problem, include the error text, current menu path, intended action, software version, and mount/camera models. Hold  and press  to save a screen image.

13 Common menu paths

Task	Menu path	Section
Focus the camera	Start(시작) → Focus(초점)	4.1
Align the eyepiece center	Start(시작) → Align(정렬) / Align (Day)(주간정렬)	4.2–4.3
Zoom the sky chart	Chart(성도) → + / –	5
Find M31	Objects(천체) → By Catalog(카탈로그별) → Messier → 3, 1 → check name → →	6.2
Search by name	Objects(천체) → Name Search(이름 검색)	6.3
Change object filters	Objects(천체) → Set Filters(필터 설정)	6.4
Log an observation	Object details → → → LOG(로그)	6.6
Measure sky brightness	SQM	7
Adjust LCD brightness	Hold □ and press + / –	3.1
Use the English LCD UI	Settings(설정) → User Pref...(사용자...) → Language(언어) → English(영어)	8.1
Change exposure / gain	Settings(설정) → Camera Exp(노출) / Camera Gain(Gain)	8.4
Connect / check the mount	Start(시작) → INDI → INIT(초기화) / STATUS(상태)	9.1–9.2
Move the mount manually	Start(시작) → INDI → Guide(가이드)	9.3
Align the mount at multiple points	Settings(설정) → INDI Setting(INDI 설정) → Multi Align(멀티 정렬)	9.4
Change automatic movement type	Settings(설정) → INDI Setting(INDI 설정) → Goto/Guide(GoTo/Guide) → GoTo Type(자동 도입 유형)	9.6
Measure lens focal length	Settings(설정) → Advanced(고급) → Lens → Auto(자동) (Measure)	10.2
Configure external input	Settings(설정) → Advanced(고급) → Bluetooth(블루투스) / Joystick(조이스틱) / Keyboard	10.5–10.6
Choose telescope / eyepiece	Tools(도구) → Equipment(관측 장비)	11.2
Enter location / time manually	Tools(도구) → Place & Time(위치/시간)	11.3
Shut down normally	Tools(도구) → Power(전원) → Shutdown(종료) → Confirm(확인)	11.7
Web access / virtual keypad	Web Home(홈) / Remote(원격 조작)	14.1–14.2
INDI MOUNT connection / detailed settings	Web INDI	14.4–14.7

14 Using the web interface

Use the web pages in a browser on a phone, tablet, or PC. You can view the LCD, use a virtual keypad, search objects, and configure the INDI MOUNT, locations, equipment, and network. The images below are captures of the current web interface; IP addresses, equipment lists, and status values are examples.

14.1 Connecting and common controls

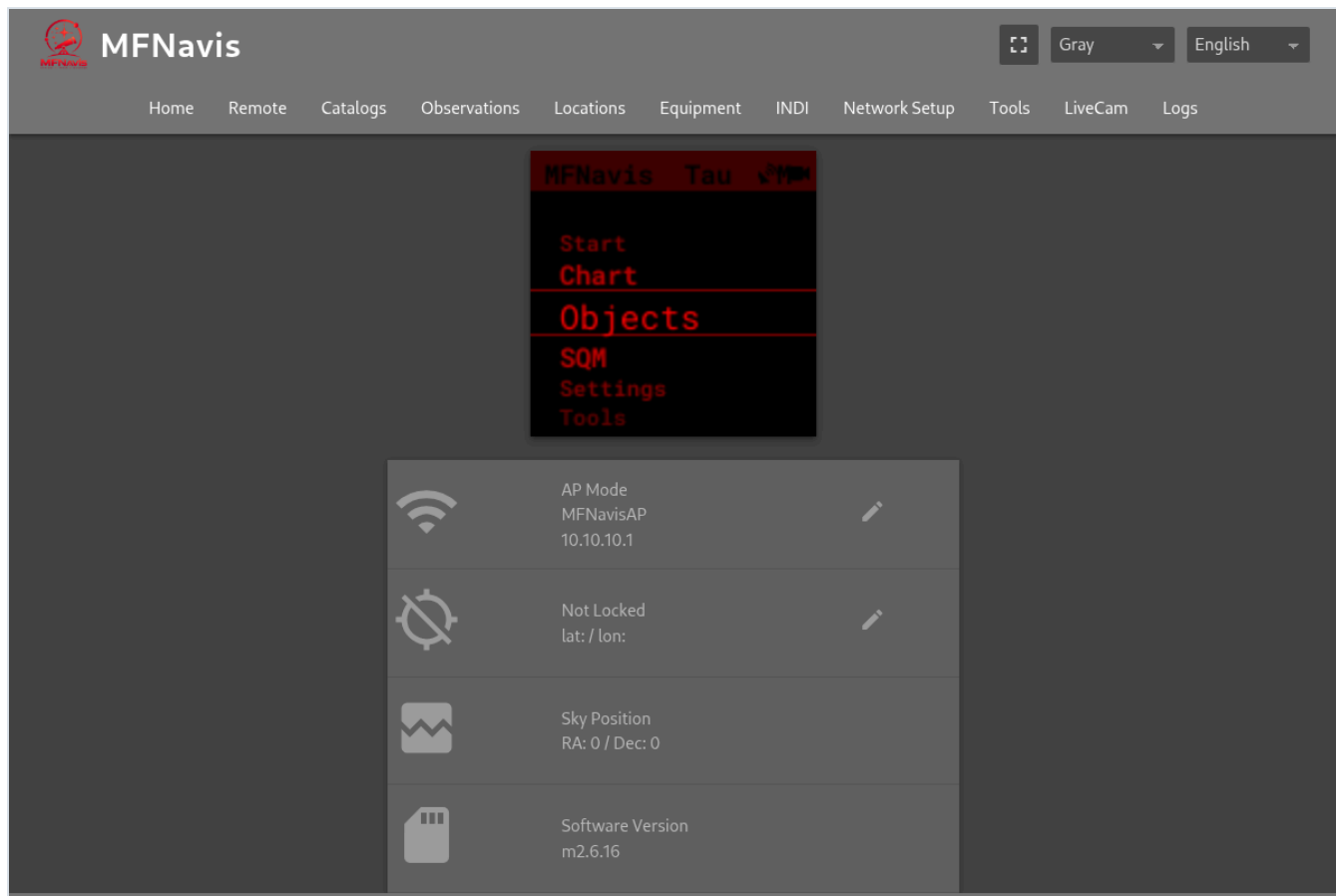
1. Connect your phone or PC to the same network as MFNavis. In the default AP configuration, join **MFNavisAP** and open **http://10.10.10.1**. Use your changed AP name or address if you have configured different values.
2. In Client(무선 연결) mode, find the device IP in LCD **Tools(도구) → Status(상태)** and open **http://device-IP**. Add **:8080** if a development server is running on port 8080.
3. If a settings or control page opens Login(로그인), enter the configured device password. Web login uses the password of the system account running the MFNavis service.
4. Choose a page from the top navigation. On phones, open ☰ for the same menu. Select English(영어) or Korean(한국어) in the language selector. Web and LCD languages are configured separately.
5. Use the save or apply button for the section you edited, then check its result message and current values. On the INDI page, **each settings section has its own apply button**.

The top bar also provides fullscreen and web-theme controls. The footer's **User manual(사용자 매뉴얼)** links open Korean or English editions, including before login. The manual's language selector keeps the current chapter.

Web menu	Purpose	Instructions
Home(홈)	LCD screen, network, GPS, coordinates, and version	14.2
Remote(원격 조작)	Virtual keypad beside the screen	14.2
Catalogs(천체 목록)	Search, filters, object details, and sending targets	14.3
Observations(관측 기록)	View and download observing sessions	14.11
Locations(관측지)	Register, edit, and load observing locations	14.8
Equipment(관측 장비)	Register and select telescopes and eyepieces	14.9
INDI	INDI MOUNT connection, controls, and detailed settings	14.4–14.7
Network Setup(네트워크 설정)	AP, Client, AP+STA, and Wi-Fi settings	14.10
Tools(도구)	Password, backup, and restore	14.12
LiveCam(실시간 영상)	Camera preview, exposure, gain, and stacking	14.13
Logs(로그)	View and download operating logs	14.12

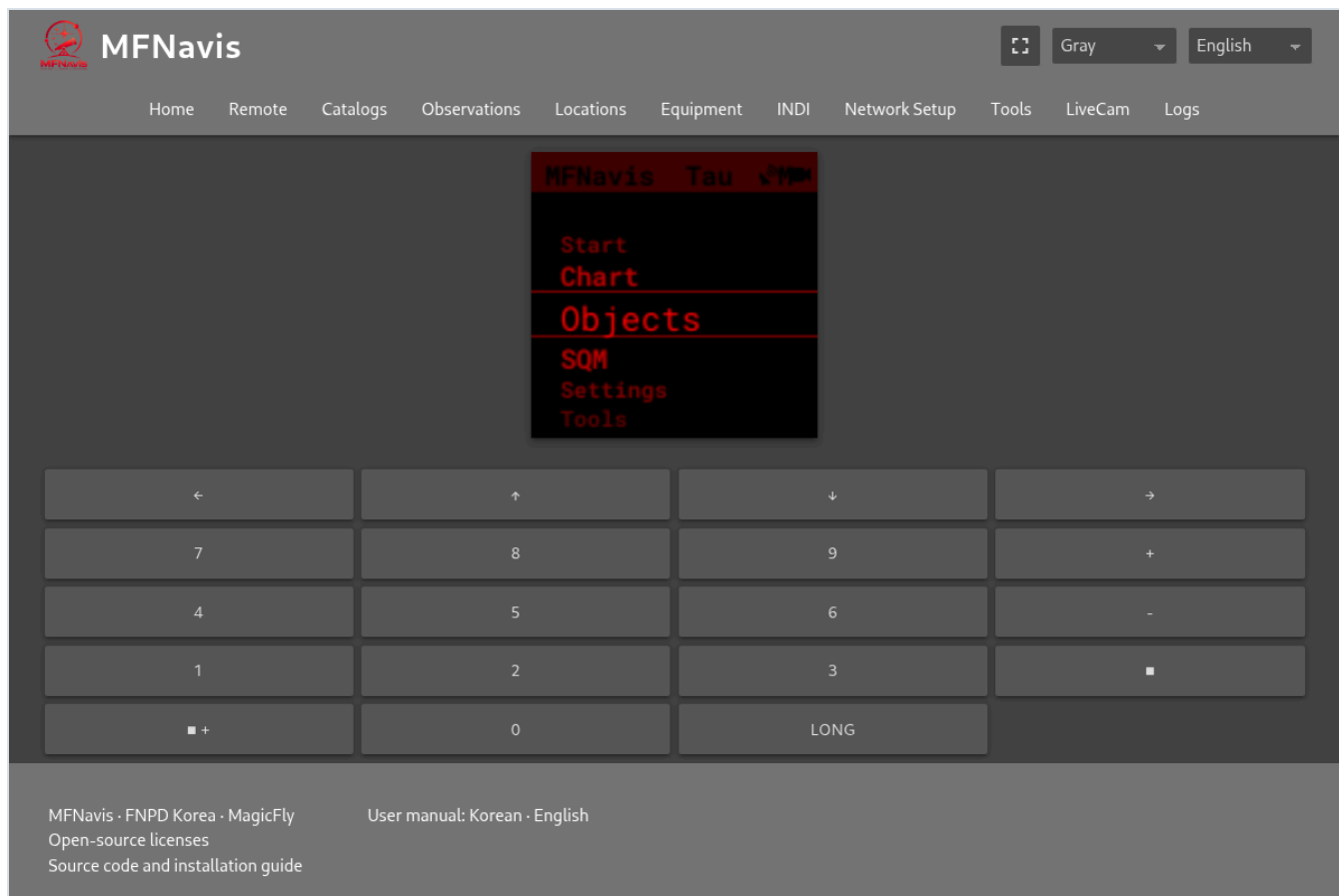
14.2 Home(홈) and Remote(원격 조작)

Path: Web navigation [Home\(홈\)](#) or [Remote\(원격 조작\)](#)



Home(홈) · LCD screen and device status

Home(홈) shows the device's current LCD image. The status table below it shows network address, GPS location, pointing coordinates, and software version. Edit icons beside network and GPS values open their settings pages.



Remote(원격 조작) · virtual keypad

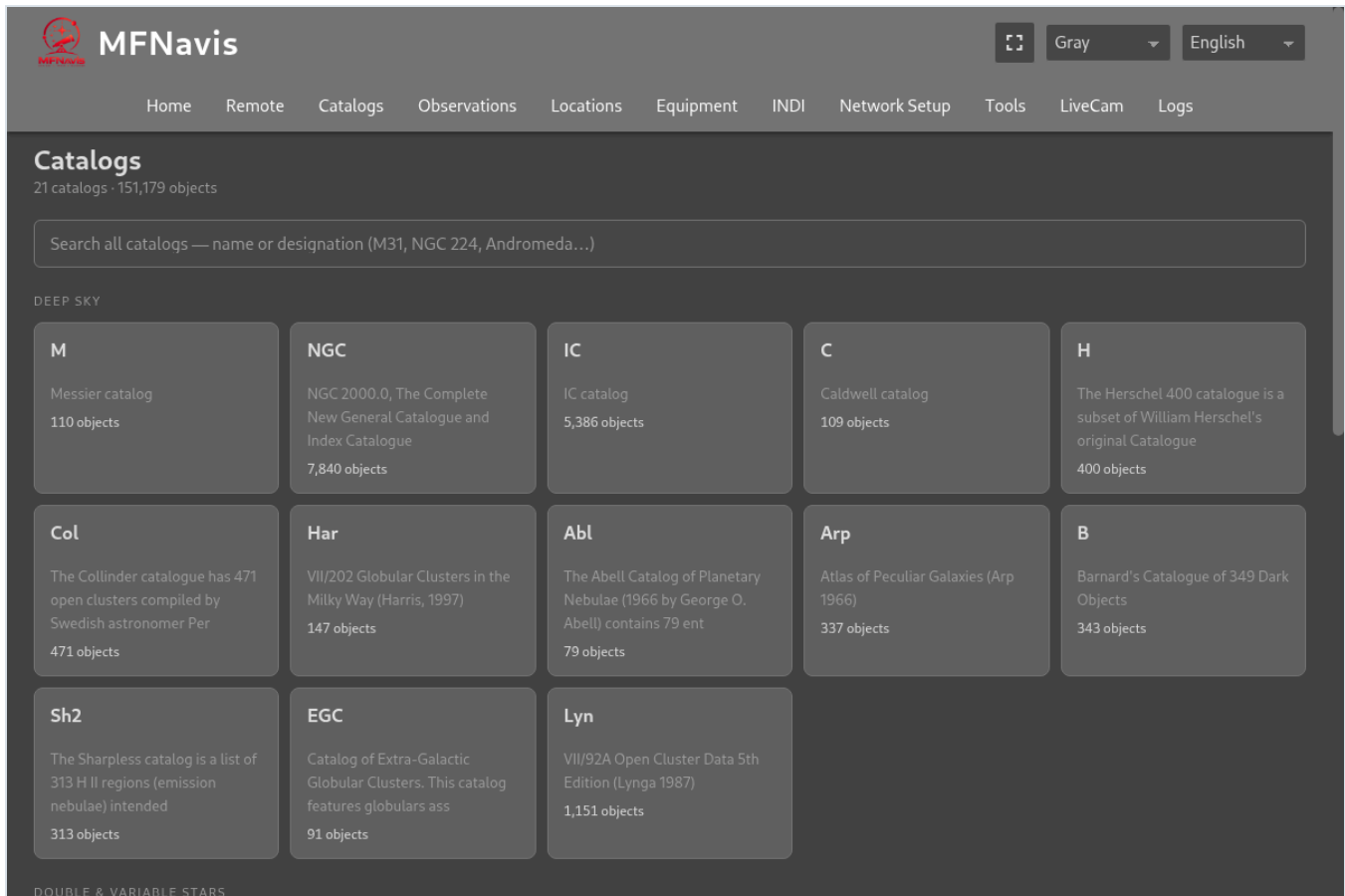
The arrows, numbers, **■**, and **+ / -** on Remote(원격 조작) send the corresponding LCD keypad button. Follow chapters 3–11 for the active LCD screen's controls.

Action	Virtual keypad sequence
Short press	Click or tap the desired key once
Long press	Select Long(경도) , then press the desired key. Example: Long(경도) → ←
Square-button combination	Select ■ + , then press the desired key. Example: ■ + → + or – adjusts LCD brightness
Cancel a modifier	Click the selected Long(경도) or ■ + again

Long(경도) and **■ +** apply to the next key once, then clear. The current Korean UI labels Long as ‘경도’; on this keypad it selects a **long press**. Select it again for another modified key. Number keys such as **5** and **0** may execute mount commands depending on the active LCD screen; check that screen first.

14.3 Catalogs(천체 목록): search and send targets

Path: Web Catalogs(천체 목록)



Catalogs(천체 목록) · global search and catalog selection

1. Enter **M31** , **NGC 224** , or a name in the global search and select a result. You can also open a catalog card.
2. Filter a catalog by name, type, constellation, magnitude, and observation status. **Up now**(현재 지평선 위) requires location information and a supported catalog. **Nearby**(주변) sorts by distance from the current pointing. Choose sorting by number, magnitude, or altitude where available.
3. Open an object to view its coordinates, magnitude, description, and altitude information. **Push to MFNavis**(MFNavis로 천체 보내기) sends it to the device's object screen.
4. With Mount Control(가대 제어) enabled and GoTo Type(자동 도입 유형) set to a value other than Off(꺼짐), sending a target can **also request automatic movement**. Check movement status and error. Use **Stop GoTo**(자동 도입 중지) to stop it. During multi-point alignment, movement follows the alignment workflow.

Returning to Catalogs(천체 목록) may resume your previous catalog or object. Open the page's Catalogs(천체 목록) breadcrumb or `/catalogs?home=1` for the catalog home. Use web observation marking and the LCD's detailed LOG(로그) entry according to the record you want to keep.

14.4 INDI MOUNT connection settings

Path: Web INDI → Current INDI Driver State(현재 INDI 드라이버 상태) / LX200 OnStepX Driver Connection(LX200 OnStepX 드라이버 연결)

LX200 OnStepX Driver Connection

Connection Type
Network TCP

INDI Server Host
localhost

INDI Server Port
7624

OnStep Network Device
Manual entry

OnStep TCP Port
9999

Manual IP or Host

Serial Auto Discovery: idle

APPLY TO INDI

INDI · LX200 OnStepX Driver Connection(LX200 OnStepX 드라이버 연결)

1. Enable LCD Tools(도구) → Experimental(실험적) → Mount Control(가대 제어) → On(켜짐) and wait for MFNavis to restart.
2. Open **INDI Web Manager(INDI 웹 관리)** on the web INDI page to select and start the profile and driver. This link opens port 8624 on the same device. Check the driver name and running state on the original INDI page.
3. Enter **INDI Server Host(INDI 서버 호스트)** and **INDI Server Port(INDI 서버 포트)**. The local defaults are localhost and 7624 . These identify the INDI server, separately from the mount's own address and port.
4. Choose **Connection Type(연결 방식)** and enter the transport settings below.
5. Click **Apply to INDI(INDI에 적용)**. Check the result and driver state. If needed, request LCD INIT(초기화) → Connect(연결) and check STATUS(상태) .

Connection	Fields	Check
Network TCP(네트워크 TCP)	Choose OnStep Network Device(OnStep 네트워크 장치), or use Manual IP or Host(수동 IP 또는 호스트). Enter OnStep TCP Port(OnStep TCP 포트)	Device choices come from AP clients. Change the default TCP port 9999 if your mount uses another port
USB Serial(USB 시리얼)	Choose USB Serial Port(USB 시리얼 포트) and Communication Speed(통신 속도), or enter a port manually	Connect the cable and reload the page to refresh the port list
USB automatic discovery	Select Auto (Find connected OnStep)(자동 (연결된 OnStep 찾기)) and, if needed, Auto (Detect with port)(자동 (포트와 함께 속도 찾기)), then apply	Requires a local INDI server, running OnStepX profile, and enabled Mount Control. Check discovery messages and the final connection result

OnStepX-specific connection, location-transfer, and mount-limit controls use the **LX200 OnStepX** driver. Other drivers may disable or omit support for these actions. **Restart INDI**(**INDI 재시작**) restarts the driver; **REBOOT INDI**(**INDI 재시작**) requests a reboot of the connected OnStep controller. Their Korean labels match, so distinguish them by their English button names.

14.5 INDI location, time, and manual movement

Path: Web **INDI** → **Location and Time**(위치 및 시간) / **Mount Control**(가대 제어)

1. Use **Reload Current Values**(현재 값 다시 읽기) to read MFNavis location and UTC time. Check latitude, longitude, and elevation, and edit the location if needed.
2. Click **Send Location and Time**(위치 및 시간 전송) and check progress and the result. The command uses MFNavis's current UTC time; the displayed time field is read-only. To change MFNavis location or time first, use **GPS Settings**(GPS 설정) or the LCD location/time menu (14.8, 11.3).
3. Check Home and Park states under **Mount Control**(가대 제어). Select **Manual Slew Rate**(수동 이동 속도), then **hold a direction button** to move. Releasing it stops manual movement.
4. Choose **Pulse Guide Rate**(펄스 가이드 속도) and click **Save**(저장) for fine corrections. This value is separate from manual slew rate.

At Home(홈 위치) sets the current Home position; Return Home(홈으로 복귀) moves there. Park(파크) moves to the park position, Unpark(언파크) releases parking, and Set-Park(파크 위치 설정) sets the current park position. Check the telescope's surroundings before requesting movement. **Reset Pointing**(좌표 초기화) under Pointing Coordinate Service(지향 좌표 서비스) rebuilds the coordinate reference.

14.6 INDI mount limits, alignment, and backlash

Path: Web **INDI** → **Settings**(설정)

Section	Action	Verify
Mount Limits(마운트 제한)	Enter overhead, horizon, and east/west meridian limits; click Apply Mount Limits (마운트 제한 적용)	Read back current values. Overhead: 60–90°; horizon: –30–30°; meridian: –180–180 minutes. Four meridian minutes equal 1°; meridian limits apply to German equatorial mounts
Multi-Point Align(다지점 정렬)	Choose Align Mode (정렬 모드), Align Points (정렬점 수) from 1–9, and Alignment Star (정렬 별) for manual mode; click Start Align (정렬 시작)	Current alignment point and status messages
Backlash(백래시)	Check axis names, enter corrections from 0–3600, and click Save Backlash (백래시 저장)	Current backlash and save result

For manual multi-point alignment, select a star, click **GoTo Selected Star**(선택한 별로 자동 도입), center it, and click **Confirm Point**(정렬점 확정). Repeat for the next point. In automatic mode, follow status messages. Use **Cancel Align**(정렬 취소) to cancel.

Backlash **Start Motion Test**(이동 테스트 시작) performs actual round-trip GoTo movement. Check the repeat count, instructions, and results. Use **Continue Motion Test**(이동 테스트 계속) when the test is waiting. Distinguish recommended candidates from stored values and confirm your chosen values with Save Backlash(백래시 저장). Axis labels follow the mount type, such as RA/DEC or AZ/ALT.

Mount-limit inputs and their apply button are disabled when driver values cannot be read. In Alt/Az mode, meridian values may be reported by INDI without direct readback from the controller.

14.7 INDI GoTo / Guide and SkySafari settings

Path: Web INDI → GoTo / Guide Settings(자동 도입 / Guide 설정)

GoTo / Guide Settings

When plate solving fails, MFNavis continues with native mount GoTo and tracking. A new solve after settling restarts GoTo and fine alignment, even with Tracking Guide off.

☐ Use unaligned IMU as a provisional GoTo reference (indoor test)

Without a compass or prior alignment, the IMU heading is relative. This test reference does not locate real sky targets. Turn it off after indoor testing.

GoTo Type

MFNavis

GoTo / Tracking Accuracy arcmin

1.0

☒ Tracking Guide

☒ Tracking Guide GoTo Recovery (re-slew when off target by more than 3 deg)

☒ Manual Re-target (after a manual move, track the new position instead of returning)

☐ Invert guide pulse RA/Az (WE)

☐ Invert guide pulse Dec/Alt (NS)

MFNavis GoTo: Max svnc + GoTo iterations

10

APPLY GOTO / GUIDE SETTINGS

PAUSE GUIDE

GoTo / Guide Settings(자동 도입 / Guide 설정) · detailed web settings

Setting	Purpose
GoTo Type(자동 도입 유형)	Off(꺼짐), INDI Mount(INDI 가대), or MFNavis; shares the saved LCD setting in 9.6
GoTo / Tracking Accuracy arcmin(GoTo 완료·추적 유지 허용 오차 (분각))	Shared arrival and tracking tolerance in positive arcminutes
Tracking Guide(추적 가이드)	Enable corrections using solved coordinates
Tracking Guide GoTo Recovery (re-slew when off target by more than 3 deg)(추적 가이드 자동 도입 복구 (목표에서 3° 이상 벗어나면 재슬루))	Recover large errors by requesting another GoTo
Manual Re-target (after a manual move, track the new position instead of returning)(수동 재타겟 (수동 이동 후 원위치 복귀 대신 새 위치를 추적))	Track the new direction after manual movement

Setting	Purpose
Invert guide pulse RA/Az (WE)(가이드 펄스 적경·방위각(WE) 반전) / Invert guide pulse Dec/Alt (NS)(가이드 펄스 적위·고도(NS) 반전)	Reverse the corresponding correction direction
MFNavis GoTo: Max sync + GoTo iterations(MFNavis 자동 도입: 최대 sync + 자동 도입 반복 횟수)	Select the maximum Sync / GoTo iteration count for MFNavis mode
Use unaligned IMU as a provisional GoTo reference (indoor test)(정렬 전 IMU를 GoTo 임시 기준으로 사용 (실내 테스트))	Relative heading for indoor testing. It does not locate real sky targets; turn it off after testing

Click **Apply GoTo / Guide Settings**(자동 도입 / Guide 설정 적용) after editing and check the result. **Pause Guide**(가이드 일시정지) temporarily pauses correction; use the resume button shown afterward to continue. Check target, phase, error, and correction state under GoTo / Guide Status(자동 도입 / Guide 상태).

During a solving failure, MFNavis continues native mount GoTo and tracking. A new solve after settling can restart fine alignment, including when Tracking Guide is off.

Under **SkySafari Mount Mode**(SkySafari 가대 모드), choose mount code, Sync, and planet-tracking options, then click **Apply SkySafari Settings**(SkySafari 설정 적용). Check that the displayed MFNavis mount type matches your equipment. **Smooth optical tracking**(부드러운 영상 추적 보정) on the same INDI page provides mode, target, and verified equipment-profile settings, followed by save, target-center confirmation, correction start, and stop controls. Response calibration and starting correction can move the mount.

14.8 Locations(관측지) and GPS Settings(GPS 설정)

Path: Web Locations(관측지) ; edit current location/time through Home(홈)'s GPS edit icon (/gps)

The screenshot shows the MFNavis web interface. At the top, there is a navigation bar with the MFNavis logo and several menu items: Home, Remote, Catalogs, Observations, Locations, Equipment, INDI, Network Setup, Tools, LiveCam, and Log. A red error message "Unexpected token '<', '" is displayed in the top right corner. The main content area is titled "Location Management" and contains a form for adding a new location. The form includes a button labeled "+ ADD NEW LOCATION". Below this, there is a section for "Location Name" with a red error message "This field is required". The "Lookup Coordinates" section contains four dropdown menus: "Country", "State / Province", "County / District", and "City / Place". Below these, there is a checkbox labeled "Use DMS Format". The "Latitude (Decimal)" and "Longitude (Decimal)" fields have red error messages "This field is required". The "Altitude (meters)" field is also present.

Location Management(관측지 관리) · add an observing location

1. Open the add-location form and enter a name. Use **Lookup Coordinates(좌표 조회)** to choose a country, region, and place, or enter latitude, longitude, and elevation manually.
2. Choose decimal degrees or **Use DMS Format(도·분·초 형식 사용)**. Check coordinate signs for north/south and east/west. Click **Save Location(위치 저장)**.
3. Use **Load Location(위치 불러오기)** to apply a saved location. **Set as Default(기본값으로 설정)** is a separate default-location choice. Use edit and delete icons to manage entries.
4. On GPS Settings(GPS 설정), check the current coordinates, elevation, date, and UTC time, then click **Save(저장)**. **Set to Browser Date/Time(브라우저 날짜·시각으로 설정)** uses your browser's clock, so verify your phone or PC time first.

Saving a location and loading it as the current location are separate actions. To send location/time to the INDI mount as well, follow 14.5.

14.9 Equipment(관측 장비): register and select

Path: Web Equipment(관측 장비)

MFNavis

Home Remote Catalogs Observations Locations Equipment INDI Network Setup Tools LiveCam Logs

Equipment

Instruments 1 Eyepieces 3 Import from DeepskyLog

ADD NEW INSTRUMENT ADD NEW EYEPIECE

Instruments

Make	Name	Aperture	Focal Length (mm)	Obstruction %	Mount Type	Flip	Flop	Reverse Arrow A	Reverse Arrow B	Active	Actions
Generic	Dobsonian	200	1000	17	alt/az	No	No	No	No	☐	

Eyepieces

Make	Name	Focal Length (mm)	Apparent FOV	Field Stop	Active	Actions
Generic	Plossl	8	50	6.5	☐	
Generic	Plossl	11	50	9.1	☐	
Generic	Plossl	25	50	21.2	☒	

Equipment(관측 장비) · telescope and eyepiece lists

1. Click **Add new instrument**(새 망원경 추가) or **Add new eyepiece**(새 접안렌즈 추가).
2. Enter name and relevant measurements: telescope aperture and focal length, eyepiece focal length and apparent field, and other applicable fields. Check units and allowed ranges, then save.
3. Select the telescope and eyepiece to use. Check the selection marker and updated magnification/field on LCD
Tools(도구) → Equipment(관측 장비) .
4. Use edit and delete icons to manage entries. DeepskyLog import fetches equipment for the entered username from an external service and requires internet access.

14.10 Network Setup(네트워크 설정)

Path: Web Network Setup(네트워크 설정)

Network Setup(네트워크 설정) · AP and Wi-Fi settings

Task	Action
Change connection mode	Select AP, Client, or AP+STA under Wifi Mode(무선 연결 방식)
Configure AP	Enter AP name, IP, security, and password; choose AP+STA internet sharing and STA band preference as needed
Save without applying	Save Settings(설정 저장) stores the edits while retaining the current connection
Apply settings	Apply & Restart(적용 후 재시작) → confirm; reconnect using the changed address and network
Register Wi-Fi	Wifi Networks(무선 네트워크) → + → choose a nearby network or enter name/password → Save(저장)
Apply saved Wi-Fi	Adjust priorities, then use Apply Now(지금 적용) when pending changes are shown. A network's connect button requests that connection

Applying and restarting or switching Wi-Fi can disconnect your browser. Use the new AP name/IP if changed. AP Connected Devices(접속점 연결 장치) lists current AP clients and also supplies the INDI network-device choices.

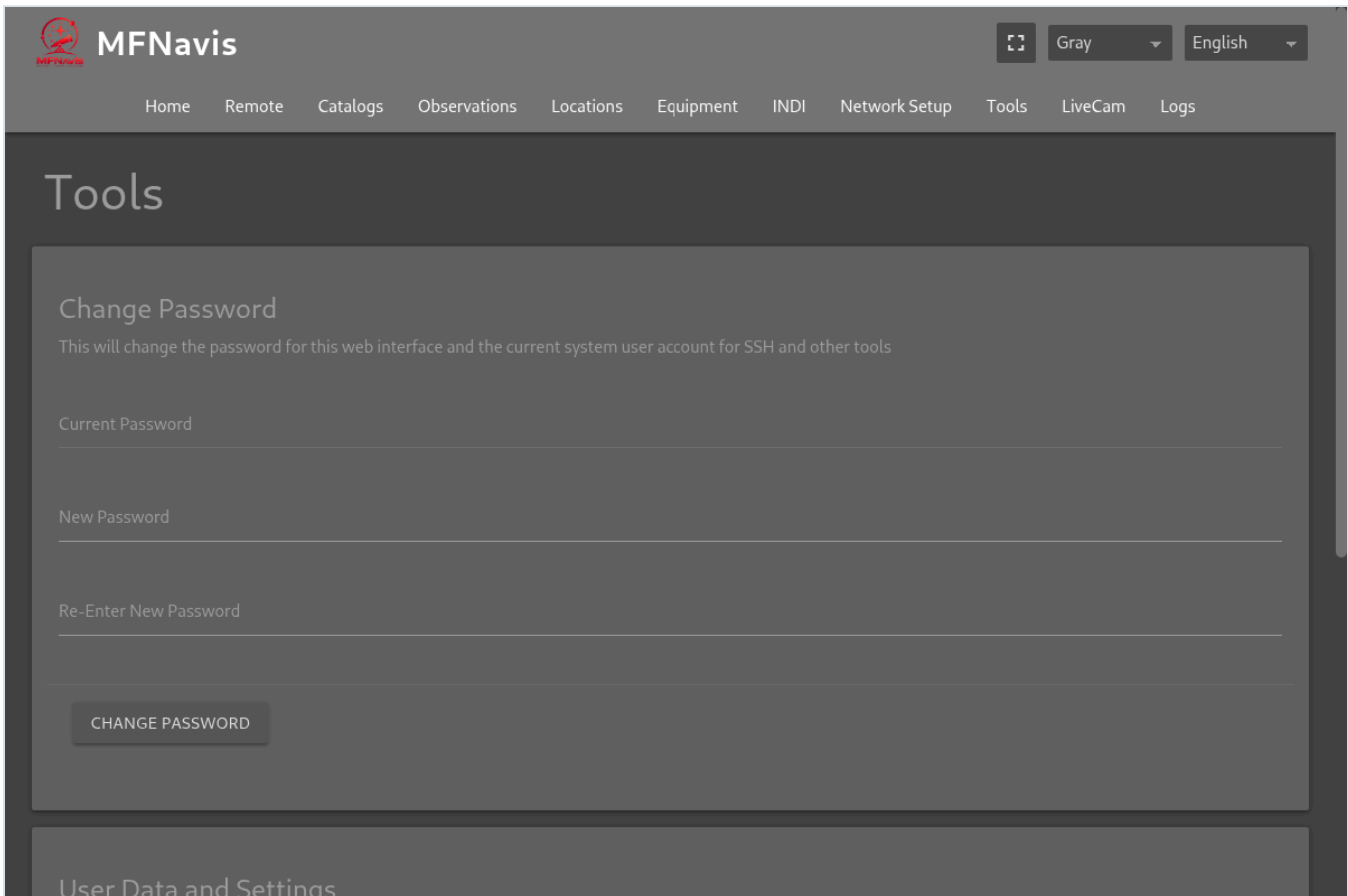
14.11 Observations(관측 기록): view and download

Path: Web Observations(관측 기록)

Select an observing session to view its objects, ratings, and notes. Download icons save all observations or the selected session as **TSV**. Open it as a tab-delimited file in a spreadsheet. See 6.6 for creating detailed observations on the LCD.

14.12 Tools(도구), Logs(로그), and diagnostic captures

Path: Web Tools(도구) or Logs(로그)



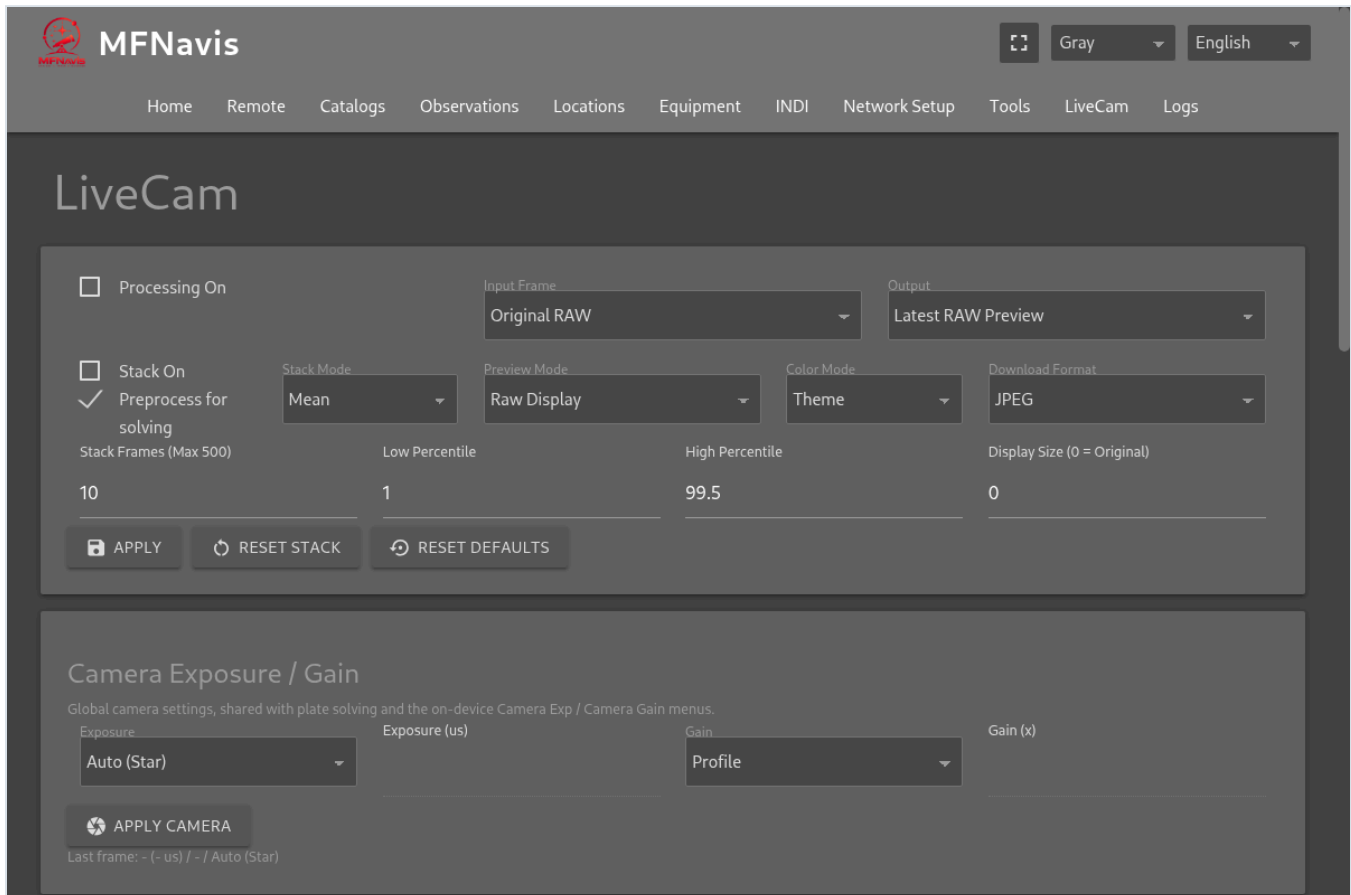
Tools(도구) · password, backup, and restore

Function	Instructions
Change Password(비밀번호 변경)	Enter current password and the new password twice. This changes both web login and that system account's SSH password
Download Backup File(백업 파일 내려받기)	Save a ZIP containing personal settings, observations, and observing lists
Upload and Restore(파일을 올려 복원)	Choose backup → restore → confirm. Existing preferences and observations are overwritten; back up current data first
Logs(로그)	Pause/resume display, copy, or use Download All Logs(모든 로그 내려받기). Save to SD(SD로 저장) saves on the device

To collect solving diagnostics, open **Plate-solving diagnostic capture**(솔빙 테스트 자료 수집) at `/solver-capture` . Choose scene, test stage, capture contents, maximum duration, attempts, and storage limit. Click **Start recording**(기록 시작) and verify the recording state. Add segment notes as needed, then click **Stop recording**(기록 종료). Recording also stops at the configured limits.

14.13 LiveCam(실시간 영상)

Path: Web `LiveCam`(실시간 영상)



LiveCam(실시간 영상) · image processing and camera controls

1. Enable **Processing On**(처리 켜짐), choose Input Frame(입력 프레임), Output(출력), and Preview Mode(미리보기 모드), then click **Apply**(적용). Choose a latest-frame preview or Live Stack(라이브 스택).
2. For stacking, choose Stack Mode(스택 모드) and frame count. Use **Reset Stack**(스택 초기화) to start a new accumulation.
3. Under **Camera Exposure / Gain**(카메라 노출 / 게인), choose exposure mode/value and gain, then click **Apply Camera**(카메라 적용). Check actual values and status. Manual exposure is in μs ; $400000\mu\text{s}$ equals 0.4 seconds.
4. Use Preview(미리보기) zoom, fit, and star overlays. Download the image in the selected format, or use TIFF download for supported RAW data.

Preprocess for solving(천구 좌표 해석용 전처리) separately affects the solver's input. Distinguish it from preview brightness, color, and stacking changes. Exposure/gain controls are unavailable when no camera process is attached.

Scope of this manual

This draft follows the MFNavis source in the working tree on October 9, 2026. Device-specific LCD displays, physical buttons, and responses from connected mounts require verification on actual hardware before release. Available entries can vary with hardware, language, saved equipment and observing lists, and software version.